NN
ACADEMY

All of IGCSE Maths in 15 Minutes

Whole-syllabus question deck — Cambridge IGCSE Mathematics 0580, Extended. One question per page, worked live on screen. Blank workspace sized for annotation in Xournal++.

58	72 / 72	15:01	16 : 9
questions	Extended subtopics	target runtime	254 × 143 mm

PREPARED FOR	DATE	SYLLABUS	EXAM SERIES
Nur Nabi	**21 September 2026**	**0580 (2025–2027) · Extended**	**2026 & 2027**

READ THIS BEFORE YOU RECORD

How to use this deck

In Xournal++

- Open this PDF as the **background** (Annotate PDF), not as an image import — the text stays crisp at any zoom.
- Page size is **254 × 143 mm**, exactly 16:9, so a full-page view fills a 1920 × 1080 recording frame with no letterboxing.
- The faint dot grid is 5 mm and is there to keep your handwriting level. It is light enough to disappear under video compression.
- Set the pen to roughly **1.5 mm**, dark blue or black. Use red only for the final answer so the freeze-frame reads at a glance.
- Export the annotated file afterwards as the downloadable "worked solutions" PDF — that is your description link and your email-list magnet.

Pacing

- Each page carries its own time budget in the top-right. It is a ceiling, not a target.
- If a page overruns, cut the explanation, never the worked answer.
- Hard cut between pages. No transitions and no "now let's look at" — there is no runtime for connective tissue.

Everything here is original. The questions were written for this deck against the published syllabus. Cambridge past-paper questions are copyright and must not be screen-recorded or retyped — these are fresh questions in the same style, which is why you can publish and monetise the video safely. The whole-syllabus sweep format is a genre, not anyone's property; the wording, the numbers and the working are yours.

Say this at the end, and put it in the description. A 15-minute sweep is a revision aid and a gap-finder, not a course. Students who rely on it alone will be short of practice. Four seconds of honesty buys you the comment section.

SEGMENT MAP

The 15-minute running order

> Paste the **Start** and **Segment** columns into the YouTube description as chapter markers. The **Budget** column is your edit discipline: Number and Algebra together take about 40% of the video because they are 31 of the 72 subtopics and they feed every other strand.

START	SEGMENT	CODES	BUDGET
0:00	**Intro — the promise**	–	0:30
0:30	**Number** 14 questions	E1.1 - E1.18	3:24
3:54	**Algebra and graphs** 14 questions	E2.1 - E2.13	3:08
7:02	**Coordinate geometry** 2 questions	E3.1 - E3.7	0:32
7:34	**Geometry** 5 questions	E4.1 - E4.8	1:20
8:54	**Mensuration** 4 questions	E5.1 - E5.5	1:06
10:00	**Trigonometry** 6 questions	E6.1 - E6.6	1:32
11:32	**Transformations and vectors** 3 questions	E7.1 - E7.4	0:52
12:24	**Probability** 4 questions	E8.1 - E8.4	0:40
13:04	**Statistics** 6 questions	E9.1 - E9.7	1:32
14:36	**Outro — how to use this**	–	0:25
15:01	**Total runtime**		

1

Number

E1.1 - E1.18 · 14 questions · 3:24 on the clock

Question 01

Types of number, HCF and LCM

(a) Write 168 and 180 each as a product of its prime factors.

(b) Hence find the highest common factor and the lowest common multiple of 168 and 180.

Also know: natural numbers · integers · primes · squares · cubes · rational and irrational · reciprocals

WORKING

Question 02

Sets and Venn diagrams

In a group of 40 students, 24 study Physics, 19 study Chemistry and 7 study neither.

(a) Complete the Venn diagram. **(b)** Find n($P \cap C$). **(c)** Find n($P \cap C'$).

Notation: ξ $\cup$ $\cap$ A' $\in$ $\notin$ $\subseteq$ $\varnothing$ $n(A)$

WORKING

Question 03

Powers, roots and the index laws

(a) Work out $27^{-2/3}$, giving your answer as a fraction.

(b) Simplify $(2a^3b)^4 \div 4a^5b^2$.

Recall squares to 15^2 and cubes of 1, 2, 3, 4, 5 and 10

WORKING

Question 04

Fractions, decimals and recurring decimals

(a) Write $0.4\dot{2}$ as a fraction in its simplest form.

(b) Work out $2\frac{1}{3} - 1\frac{3}{4}$, giving your answer as a fraction in its simplest form.

WORKING

Question 05

Ordering and the four operations

(a) Write these in order, smallest first: -0.7 $-\frac{3}{4}$ $-\frac{5}{8}$ -0.72

(b) Work out $-3 + 4 \times (-2)^2$.

Symbols: = ≠ > < ≥ ≤

WORKING

Question 06

Standard form

(a) Work out $(3.2 \times 10^{5}) \div (8 \times 10^{-2})$.

(b) Work out $(6 \times 10^{4}) + (4.5 \times 10^{5})$.

Answers must be in the form $A \times 10^{n}$ with $1 \leq A < 10$

WORKING

Question 07

Estimation

By writing each number correct to 1 significant figure, estimate the value of

$$(4.87 \times 10.2) \div 0.213$$

Show the rounded values — the method carries the marks

WORKING

Question 08

Limits of accuracy (bounds)

A rectangle measures 12.4 cm by 8.6 cm, each correct to 1 decimal place.

(a) Write down the lower and upper bound of each side.

(b) Calculate the upper bound of the area. **(c)** Calculate the lower bound of the perimeter.

WORKING

Question 09

Ratio and proportion

An amount of money is shared between Amina, Bilal and Chowdhury in the ratio 3 : 5 : 7.

Chowdhury receives $84 more than Amina. Find the total amount of money shared.

WORKING

Question 10

Rates

(a) A train travels 150 km in 2 hours 30 minutes. Find its average speed in km/h.

(b) A metal block has mass 4.8 kg and volume 600 cm^3. Find its density in g/cm^3.

speed = distance ÷ time · density = mass ÷ volume · pressure = force ÷ area

WORKING

Question 11

Percentages and reverse percentages

(a) After a 15% increase, the price of a phone is $161. Calculate the price before the increase.

(b) $2000 is invested at 3.5% per year compound interest. Find the value after 5 years.

WORKING

Question 12

Calculator, time and money

(a) A flight leaves at 21:45 and takes 7 hours 40 minutes. The destination is 3 hours ahead. Find the arrival time, local time.

(b) 1 US dollar = 118 taka. Convert 250 US dollars to taka.

Keep full accuracy in the calculator; round only the final answer to 3 s.f. unless told otherwise

WORKING

Question 13

Exponential growth and decay

A car is bought for $18 000. Its value depreciates by 12% each year.

Calculate its value after 4 years, correct to the nearest dollar.

Growth: $P(1 + r)^n$ · Decay: $P(1 - r)^n$ · knowledge of e is not required

WORKING

Question 14

Surds

(a) Simplify √50 + √18. **(b)** Simplify 6 ÷ √3.

(c) Rationalise the denominator of 1 ÷ (2 + √3).

WORKING

2

Algebra and graphs

E2.1 - E2.13 · 14 questions · 3:08 on the clock

Question 15

Substitution into formulas

$v = u + at$ and $s = ut + \frac{1}{2}at^2$

Find v and s when $u = 7$, $a = -2.5$ and $t = 4$.

WORKING

Question 16

Expanding and factorising

(a) Expand and simplify $(2x - 3)(x + 5)$.

(b) Factorise $6x^2 - x - 12$. **(c)** Factorise $9x^2 - 49y^2$.

(d) Factorise by grouping: $ax + ay - 2bx - 2by$.

WORKING

Question 17

Algebraic fractions

(a) Simplify $(x^2 - 9) \div (x^2 + 7x + 12)$.

(b) Write as a single fraction $3/(x + 1) + 2/(x - 2)$.

WORKING

Question 18

Indices II

(a) Solve $2^{x+1} = 32$. **(b)** Solve $3^{2x} = 81$.

(c) Simplify $(8x^6)^{2/3}$.

WORKING

Question 19

Equations and rearranging

(a) Solve simultaneously $3x + 2y = 16$ and $x - y = 2$.

(b) Solve $2x^2 - 7x - 3 = 0$, giving answers correct to 2 decimal places.

(c) Make r the subject of $V = \frac{2}{3}\pi r^3$. **(d)** Write $x^2 - 6x + 2$ in the form $(x - a)^2 - b$.

WORKING

Question 20

Non-linear simultaneous equations

Solve simultaneously

$$y = x^2 - 4 \quad \text{and} \quad y = 3x$$

WORKING

Question 21

Inequalities

(a) Solve $5 - 2x < 11$ and show the solution on a number line.

(b) Solve $-3 \leq 2x + 1 < 9$.

Dividing by a negative reverses the inequality sign

WORKING

Question 22

Sequences

Find the *n*th term of each sequence.

(a) 7, 11, 15, 19, … **(b)** 3, 8, 15, 24, 35, … **(c)** 5, 10, 20, 40, …

Linear · quadratic · cubic · exponential · and simple combinations

WORKING

Question 23

Direct and inverse proportion

y is inversely proportional to x^2. When $x = 2$, $y = 12$.

(a) Find y in terms of x. **(b)** Find y when $x = 4$. **(c)** Find x when $y = 0.75$.

WORKING

Question 24

Graphs in practical situations

A car accelerates uniformly from rest to 15 m/s in 10 s, travels at 15 m/s for 30 s, then decelerates uniformly to rest in 20 s.

(a) Sketch the speed–time graph. **(b)** Find the acceleration in the first 10 s. **(c)** Find the total distance travelled.

WORKING

Question 25

Graphs of functions

$y = x^3 - 3x + 1$ for $-2 \le x \le 2$

(a) Complete the table of values. **(b)** Draw the graph. **(c)** Use your graph to solve $x^3 - 3x + 1 = 0$.

WORKING

Question 26

Sketching curves

(a) Write $y = x^2 - 6x + 5$ in completed-square form and state the turning point and the roots.

(b) Sketch $y = 4/x$ and state the equations of its asymptotes.

Know the shapes: linear · quadratic · cubic · reciprocal · exponential

WORKING

Question 27

Differentiation

$y = x^3 - 6x^2 + 5$

(a) Find dy/dx. **(b)** Find the coordinates of the two stationary points.

(c) Determine which is a maximum and which is a minimum.

WORKING

Question 28

Functions

$f(x) = 3x - 4 \quad g(x) = x^2$

(a) Find fg(2). **(b)** Find gf(2). **(c)** Find $f^{-1}(x)$. **(d)** Find ff(x) in its simplest form.

WORKING

3

Coordinate geometry

E3.1 - E3.7 · 2 questions · 0:32 on the clock

Question 29

Coordinates, length, midpoint and equation of a line

A is the point $(-2, 1)$ and B is the point $(6, 7)$.

(a) Find the midpoint of AB. **(b)** Find the length of AB. **(c)** Find the gradient of AB.

(d) Find the equation of the line AB in the form $y = mx + c$.

WORKING

Question 30

Parallel and perpendicular lines

Using $A(-2, 1)$ and $B(6, 7)$ from the previous question:

(a) Find the equation of the line through the origin parallel to AB.

(b) Find the equation of the line through B perpendicular to AB.

Perpendicular gradients multiply to give -1

WORKING

4

Geometry

E4.1 - E4.8 · 5 questions · 1:20 on the clock

Question 31

Geometrical terms and angles

(a) For a regular 12-sided polygon, find the size of each exterior angle, each interior angle, and the sum of the interior angles.

(b) Two parallel lines are cut by a transversal. Name the angle relationships you would use and give a reason for each.

WORKING

Question 32

Constructions, scale drawings and bearings

(a) Using a ruler and compasses only, construct the perpendicular bisector of a line and the bisector of an angle. Leave your construction arcs visible.

(b) B is 8 km from A on a bearing of 125°. Find the bearing of A from B.

WORKING

Question 33

Similarity and symmetry

Two similar cones have heights 4 cm and 6 cm. The smaller cone has volume 32 cm^3 and surface area 20 cm^2.

(a) Find the volume of the larger cone. **(b)** Find its surface area.

Length factor k → area factor k^2 → volume factor k^3

WORKING

Question 34

Circle theorems I

Find the lettered angles, giving a geometrical reason for each.

(a) The angle at the centre is 130°; find the angle at the circumference on the major arc.

(b) Hence find the opposite angle of the cyclic quadrilateral.

(c) A tangent meets a chord at 58°; find the angle in the alternate segment.

Angle in a semicircle · tangent ⊥ radius · centre = 2 × circumference · same segment · cyclic quadrilateral · alternate segment

WORKING

Question 35

Circle theorems II — symmetry properties

P is a point outside a circle, centre *O* and radius 8 cm. *PT* is a tangent and *OP* = 17 cm.

(a) Find the length *PT*. **(b)** State what you know about the second tangent from *P*.

Equal chords are equidistant from the centre · the perpendicular bisector of a chord passes through the centre · tangents from an external point are equal

WORKING

5

Mensuration

E5.1 - E5.5 · 4 questions · 1:06 on the clock

Question 36

Units of measure

Convert:

(a) 2.5 m^2 into cm^2 **(b)** 3.2 m^3 into cm^3 **(c)** 750 cm^3 into litres

WORKING

Question 37

Area, perimeter and compound shapes

(a) A trapezium has parallel sides 7 cm and 11 cm and perpendicular height 6 cm. Find its area.

(b) A rectangle 10 cm by 6 cm has a semicircle of diameter 6 cm removed from one end. Find the area remaining.

WORKING

Question 38

Circles, arcs and sectors

A sector of a circle has radius 9 cm and angle 140°.

(a) Find the arc length. **(b)** Find the area of the sector. **(c)** Find the perimeter of the sector.

WORKING

Question 39

Surface area and volume

(a) A cone has radius 5 cm and vertical height 12 cm. Find its slant height, its total surface area and its volume.

(b) A sphere has radius 6 cm. Find its volume and its surface area.

Cone: $\frac{1}{3}\pi r^2 h$ and curved area $\pi r l$ · Sphere: $\frac{4}{3}\pi r^3$ and $4\pi r^2$ — both given in the formula list

WORKING

6

Trigonometry

E6.1 - E6.6 · 6 questions · 1:32 on the clock

Question 40

Pythagoras and right-angled trigonometry

In a right-angled triangle the side opposite angle θ is 7 cm and the hypotenuse is 12 cm.

(a) Find θ. **(b)** Find the third side.

SOH · CAH · TOA

WORKING

Question 41

Exact trigonometric values

Complete the table of exact values, without a calculator.

	0°	30°	45°	60°	90°
sin					
cos					
tan					—

WORKING

Question 42

Trigonometric graphs and equations

Solve for $0° \leq x \leq 360°$.

(a) $\sin x = 0.5$ **(b)** $2\cos x + 1 = 0$

Sketch the curve first — it tells you how many solutions there are

WORKING

Question 43

The sine rule

In triangle ABC, angle $A = 47°$, angle $B = 63°$ and side $a = 8.4$ cm.

Find the length of side b.

$a / \sin A = b / \sin B$ — given in the formula list. Watch for the ambiguous case.

WORKING

Question 44

The cosine rule and area of a triangle

In triangle ABC, $b = 8$ cm, $c = 11$ cm and angle $A = 47°$.

(a) Find the length a. **(b)** Find the area of the triangle.

$a^2 = b^2 + c^2 - 2bc\cos A$ · $\text{area} = \frac{1}{2}ab\sin C$

WORKING

Question 45

Pythagoras and trigonometry in 3D

A cuboid has a rectangular base 6 cm by 8 cm and height 10 cm.

(a) Find the length of the space diagonal.

(b) Find the angle between the space diagonal and the base.

WORKING

7

Transformations and vectors

E7.1 - E7.4 · 3 questions · 0:52 on the clock

Question 46

Transformations

Describe each transformation fully.

(a) A reflection. **(b)** A rotation. **(c)** An enlargement with scale factor −2, centre (1, 1). **(d)** A translation.

Reflection: the mirror line · Rotation: angle, direction, centre · Enlargement: scale factor, centre · Translation: the column vector

WORKING

Question 47

Vectors and magnitude

$\mathbf{a} = (3, -1)$ and $\mathbf{b} = (-2, 5)$, written as column vectors.

(a) Find $2\mathbf{a} + \mathbf{b}$. **(b)** Find $|2\mathbf{a} + \mathbf{b}|$. **(c)** Find $\mathbf{a} - \mathbf{b}$.

WORKING

Question 48

Vector geometry

$OA = \mathbf{a}$ and $OB = \mathbf{b}$. M is the midpoint of AB.

(a) Find AB in terms of $\mathbf{a}$ and $\mathbf{b}$. **(b)** Find OM.

(c) N divides OB in the ratio 2 : 1. Find AN.

WORKING

8

Probability

E8.1 - E8.4 · 4 questions · 0:40 on the clock

Question 49

Introduction to probability

A spinner has 8 equal sectors: 3 red, 2 blue and 3 green.

(a) Find P(red). **(b)** Find P(not blue). **(c)** Explain why P(yellow) = 0.

All probabilities lie between 0 and 1 · P(A') = 1 − P(A)

WORKING

Question 50

Relative and expected frequency

The probability that a machine produces a faulty component is 0.15.

(a) Find the expected number of faulty components in a batch of 400.

(b) In a trial, 72 out of 400 were faulty. Find the relative frequency and comment.

WORKING

Question 51

Combined events and tree diagrams

A bag contains 5 red and 3 blue counters. Two counters are taken at random **without replacement**.

(a) Draw a tree diagram. **(b)** Find P(both red). **(c)** Find P(one of each colour).

AND → multiply along branches · OR → add the branch outcomes

WORKING

Question 52

Conditional probability

Using the same bag of 5 red and 3 blue counters, taken without replacement:

(a) Find the probability that the second counter is red, given the first was red.

(b) Find the probability that the second is blue, given the first was red.

WORKING

9

Statistics

E9.1 - E9.7 · 6 questions · 1:32 on the clock

Question 53

Classifying and interpreting data

Classify each as qualitative or quantitative, and where quantitative, as discrete or continuous.

(a) Eye colour **(b)** Number of siblings **(c)** Height **(d)** Shoe size

WORKING

Question 54

Averages and measures of spread

The table shows the time t minutes taken by 30 students to finish a task.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$
Frequency	4	11	9	6

(a) Calculate an estimate of the mean. **(b)** Write down the modal class. **(c)** Identify the class containing the median.

WORKING

Question 55

Statistical charts and diagrams

(a) In a survey of 30 students, 11 chose football. Find the angle for football on a pie chart.

(b) Draw an ordered stem-and-leaf diagram for: 34, 21, 45, 27, 38, 41, 22, 36.

Bar charts (including stacked and dual) · pie charts · pictograms · stem-and-leaf — a stem-and-leaf must have a key

WORKING

Question 56

Scatter diagrams and correlation

A scatter diagram shows hours of revision against test score.

(a) Describe the type of correlation you would expect. **(b)** Draw a line of best fit. **(c)** Use it to estimate the score for 6 hours of revision.

A line of best fit is a single ruled line, drawn by inspection, across the full data set

WORKING

Question 57

Cumulative frequency diagrams

A cumulative frequency curve is drawn for 80 values.

(a) At what value of the cumulative frequency do you read the median, the lower quartile and the upper quartile?

(b) If LQ = 24 and UQ = 41, find the interquartile range.

*Plot points at the **upper** class boundary and join with a smooth curve*

WORKING

Question 58

Histograms

A histogram has unequal class widths.

(a) The class $20 < x \leq 50$ has frequency 24. Find the frequency density.

(b) Another bar has frequency density 2.4 over the class $50 < x \leq 60$. Find its frequency.

*frequency density = frequency ÷ class width — the vertical axis is **always** labelled Frequency density*

WORKING

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Answer key 1 of 6

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
01	**E1.1**	Types of number, HCF and LCM	$168 = 2^3 \times 3 \times 7$ $180 = 2^2 \times 3^2 \times 5$ | HCF $= 2^2 \times 3 =$ **12** LCM $= 2^3 \times 3^2 \times 5 \times 7 =$ **2520**
02	**E1.2**	Sets and Venn diagrams	$n(P \cup C) = 40 - 7 = 33$, so $24 + 19 - x = 33 \rightarrow$ **$n(P \cap C) = 10$**; $n(P \cap C') = 24 - 10 =$ **14**
03	**E1.3 / E1.7**	Powers, roots and the index laws	(a) $27^{-2/3} = 1 \div (\sqrt[3]{27})^2 = 1 \div 9 =$ **1/9** | (b) $16a^{12}b^4 \div 4a^5b^2 =$ **$4a^7b^2$**
04	**E1.4**	Fractions, decimals and recurring decimals	(a) $0.4\dot{2} = 42/99 =$ **14/33** | (b) $7/3 - 7/4 = (28 - 21)/12 =$ **7/12**
05	**E1.5 / E1.6**	Ordering and the four operations	(a) −0.75, −0.72, −0.7, −0.625 → **−¾, −0.72, −0.7, −⅝** | (b) $-3 + 4 \times 4 =$ **13**
06	**E1.8**	Standard form	(a) $0.4 \times 10^7 =$ **4×10^6** | (b) 60 000 + 450 000 = 510 000 = **5.1×10^5**
07	**E1.9**	Estimation	$(5 \times 10) \div 0.2 = 50 \div 0.2 =$ **250**
08	**E1.10**	Limits of accuracy (bounds)	$12.35 \le L < 12.45$, $8.55 \le W < 8.65$ | (b) $12.45 \times 8.65 =$ **107.6925 cm²** | (c) $2(12.35 + 8.55) =$ **41.8 cm**
09	**E1.11**	Ratio and proportion	7 − 3 = 4 parts = \$84, so 1 part = \$21. Total = 15 × 21 = **\$315**
10	**E1.12**	Rates	(a) 150 ÷ 2.5 = **60 km/h** | (b) 4800 g ÷ 600 cm³ = **8 g/cm³**

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Answer key 2 of 6

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
11	E1.13	Percentages and reverse percentages	(a) 161 ÷ 1.15 = **$140** (not 161 × 0.85) | (b) 2000 × 1.035^5 = **$2375.37**
12	E1.14 / E1.15 / E1.16	Calculator, time and money	(a) 21:45 + 7 h 40 = 05:25 next day, + 3 h = **08:25** | (b) 250 × 118 = **29 500 taka**
13	E1.17	Exponential growth and decay	18 000 × 0.88^4 = **$10 795** (10 794.55)
14	E1.18	Surds	(a) 5√2 + 3√2 = **8√2** | (b) 6√3/3 = **2√3** | (c) × (2 − √3)/(2 − √3) → (2 − √3)/1 = **2 − √3**
15	E2.1	Substitution into formulas	v = 7 + (−2.5)(4) = **−3** | s = 7(4) + ½(−2.5)(16) = 28 − 20 = **8**
16	E2.2	Expanding and factorising	(a) **$2x^2 + 7x - 15$** | (b) **(2x − 3)(3x + 4)** | (c) **(3x − 7y)(3x + 7y)** | (d) **(a − 2b)(x + y)**
17	E2.3	Algebraic fractions	(a) (x − 3)(x + 3) / (x + 3)(x + 4) = **(x − 3)/(x + 4)** | (b) **(5x − 4) / (x + 1)(x − 2)**
18	E2.4	Indices II	(a) x + 1 = 5 → **x = 4** | (b) 2x = 4 → **x = 2** | (c) **$4x^4$**
19	E2.5	Equations and rearranging	(a) **x = 4, y = 2** | (b) x = (7 ± √73)/4 = **3.89 or −0.39** | (c) **r = $\sqrt[3]{(3V / 4\pi)}$** | (d) **$(x - 3)^2 - 7$**
20	E2.5	Non-linear simultaneous equations	x^2 − 3x − 4 = 0 → (x − 4)(x + 1) = 0 → **x = 4, y = 12** or **x = −1, y = −3**

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Answer key 3 of 6

Check these before you record. Every value has been worked through; rounding is to 3 significant figures unless the question says otherwise.

	CODE	TOPIC	ANSWER
21	E2.6	Inequalities	(a) $-2x < 6 \rightarrow$ **$x > -3$** (open circle) | (b) $-4 \le 2x < 8 \rightarrow$ **$-2 \le x < 4$**
22	E2.7	Sequences	(a) **$4n + 3$** | (b) second difference 2 → **$n^2 + 2n$** | (c) **$5 \times 2^{n-1}$**
23	E2.8	Direct and inverse proportion	(a) **$y = 48/x^2$** | (b) **$y = 3$** | (c) $x^2 = 64 \rightarrow$ **$x = 8$**
24	E2.9	Graphs in practical situations	(b) $15 \div 10 =$ **1.5 m/s²** | (c) area = ½(10)(15) + 30(15) + ½(20)(15) = 75 + 450 + 150 = **675 m**
25	E2.10	Graphs of functions	Table: −1, 3, 1, −1, 3 at x = −2, −1, 0, 1, 2 | roots ≈ **−1.88, 0.35, 1.53**
26	E2.11	Sketching curves	(a) $(x - 3)^2 - 4$, turning point **(3, −4)**, roots **x = 1 and x = 5** | (b) asymptotes **x = 0 and y = 0**
27	E2.12	Differentiation	(a) **$3x^2 - 12x$** | (b) $3x(x - 4) = 0 \rightarrow$ **(0, 5)** and **(4, −27)** | (c) $d^2y/dx^2 = 6x - 12$: at x = 0 it is −12 → **maximum**; at x = 4 it is 12 → **minimum**
28	E2.13	Functions	(a) $f(4) =$ **8** | (b) $g(2) =$ **4** | (c) **$(x + 4)/3$** | (d) $3(3x - 4) - 4 =$ **$9x - 16$**
29	E3.1 - E3.5	Coordinates, length, midpoint and equation of a line	(a) **(2, 4)** | (b) $\sqrt{(8^2 + 6^2)} =$ **10** | (c) 6/8 = **¾** | (d) **$y = ¾x + 2.5$**
30	E3.6 / E3.7	Parallel and perpendicular lines	(a) **$y = ¾x$** | (b) gradient = $-1 \div ¾ = -4/3$, through (6, 7) → **$y = -(4/3)x + 15$**

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Answer key 4 of 6

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	CODE	TOPIC	ANSWER
31	E4.1 / E4.6	Geometrical terms and angles	(a) exterior = 360 ÷ 12 = **30°**; interior = **150°**; sum = (12 − 2) × 180 = **1800°** | (b) corresponding, alternate, co-interior (sum 180°)
32	E4.2 / E4.3	Constructions, scale drawings and bearings	(b) 125 + 180 = **305°** (bearings are always given as three figures, measured clockwise from north)
33	E4.4 / E4.5	Similarity and symmetry	k = 1.5 | (a) 32 × 1.5^3 = 32 × 3.375 = **108 cm³** | (b) 20 × 1.5^2 = **45 cm²**
34	E4.7	Circle theorems I	(a) **65°** (angle at centre is twice the angle at the circumference) | (b) 180 − 65 = **115°** (opposite angles of a cyclic quadrilateral) | (c) **58°** (alternate segment theorem)
35	E4.8	Circle theorems II — symmetry properties	(a) $\sqrt{(17^2 - 8^2)}$ = √225 = **15 cm** (tangent ⊥ radius) | (b) it is also **15 cm** — tangents from an external point are equal in length
36	E5.1	Units of measure	(a) 2.5 × 10 000 = **25 000 cm²** | (b) 3.2 × 1 000 000 = **3 200 000 cm³** | (c) **0.75 litres**
37	E5.2 / E5.5	Area, perimeter and compound shapes	(a) ½(7 + 11)(6) = **54 cm²** | (b) 60 − ½π$(3)^2$ = 60 − 14.14 = **45.9 cm²**
38	E5.3	Circles, arcs and sectors	(a) (140/360)(2π × 9) = **22.0 cm** | (b) (140/360)(π × 81) = **99.0 cm²** | (c) 21.99 + 9 + 9 = **40.0 cm**
39	E5.4	Surface area and volume	(a) l = 13 cm; total SA = π(5)(13) + π$(5)^2$ = 90π = **283 cm²**; V = ⅓π(25)(12) = 100π = **314 cm³** | (b) V = 288π = **905 cm³**; SA = 144π = **452 cm²**
40	E6.1 / E6.2	Pythagoras and right-angled trigonometry	(a) sin θ = 7/12 → **θ = 35.7°** | (b) √(144 − 49) = √95 = **9.75 cm**

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Answer key 5 of 6

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	CODE	TOPIC	ANSWER
41	E6.3	Exact trigonometric values	sin: 0, ½, √2/2, √3/2, 1 | cos: 1, √3/2, √2/2, ½, 0 | tan: 0, 1/√3, 1, √3 (tan 90° undefined)
42	E6.4	Trigonometric graphs and equations	(a) **x = 30° and 150°** | (b) cos x = −0.5 → **x = 120° and 240°**
43	E6.5	The sine rule	b = 8.4 × sin 63° ÷ sin 47° = **10.2 cm**
44	E6.5	The cosine rule and area of a triangle	(a) a^2 = 64 + 121 − 176 cos 47° = 64.97 → **a = 8.06 cm** | (b) ½(8)(11) sin 47° = **32.2 cm^2**
45	E6.6	Pythagoras and trigonometry in 3D	(a) √(36 + 64 + 100) = √200 = **14.1 cm** | (b) base diagonal = 10, tan θ = 10/10 → **θ = 45°**
46	E7.1	Transformations	A negative scale factor puts the image on the **opposite side of the centre and inverted**; a fractional scale factor makes it **smaller**. One word per transformation — naming two transformations loses the mark.
47	E7.2 / E7.3	Vectors and magnitude	(a) (6, −2) + (−2, 5) = **(4, 3)** | (b) √(16 + 9) = **5** | (c) **(5, −6)**
48	E7.4	Vector geometry	(a) **b − a** | (b) **a** + ½(**b** − **a**) = **½(a + b)** | (c) **⅔b − a**
49	E8.1	Introduction to probability	(a) **3/8** | (b) 1 − 2/8 = **3/4** | (c) impossible outcome
50	E8.2	Relative and expected frequency	(a) 0.15 × 400 = **60** | (b) 72/400 = **0.18** — higher than expected, though a single trial is weak evidence

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Answer key 6 of 6

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	CODE	TOPIC	ANSWER
51	E8.3	Combined events and tree diagrams	(b) 5/8 × 4/7 = 20/56 = **5/14** | (c) 2 × (5/8 × 3/7) = 30/56 = **15/28**
52	E8.4	Conditional probability	(a) **4/7** | (b) **3/7** — the condition changes the denominator as well as the numerator
53	E9.1 / E9.2	Classifying and interpreting data	(a) qualitative | (b) quantitative, **discrete** | (c) quantitative, **continuous** | (d) quantitative, discrete
54	E9.3	Averages and measures of spread	(a) Σfx = 20 + 165 + 225 + 210 = 620; 620 ÷ 30 = **20.7 min** | (b) **10 < t ≤ 20** | (c) 15th/16th value → **20 < t ≤ 30**
55	E9.4	Statistical charts and diagrams	(a) 11/30 × 360 = **132°** | (b) 2 | 1 2 7 3 | 4 6 8 4 | 1 5 **Key: 2 | 1 means 21**
56	E9.5	Scatter diagrams and correlation	(a) **positive correlation** | (b) roughly even numbers of points either side, full width | (c) read off the line, not the nearest point
57	E9.6	Cumulative frequency diagrams	(a) median at **40**, LQ at **20**, UQ at **60** | (b) IQR = 41 − 24 = **17**
58	E9.7	Histograms	(a) 24 ÷ 30 = **0.8** | (b) 2.4 × 10 = **24**